Name : V. Smylika

SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: Theory of Computation **COURSE CODE:** CSA13

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

CLASS TEST

Code of Conduct:

I V. Smylika (192372115) *(Name / Reg No) certify that this submission is my original work and that I ha*
adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Smylika.V.

Signature of the student:

SIMATS ENGINEERING
ASSESSMENT TOOLS

1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b' where 'n' is a non-negative integer.
i) $L = \{a^n b^n\}$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.
$E \rightarrow E+E \mid E*E \mid (E) \mid I$
$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

3. Design a TM to accepts the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b' where 'n' is a non-negative integer
i) $L = \{0^n 1^n \mid n \geq 1\}$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice than that of a's.

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

5. Construct a Turing Machine to recognize the language.Write the logic used for the design and the formal definition

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

Class Test-02.
CO4 Assessment Tool 3

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Given:- $L = \{a^n b^n\}$

PDA:

The PDA uses its stack to count the number of as.

- For every A, push one A onto the stack.
- When Bs begin, pop one A for every B
- Accept when all A's have been poppoed and the input is accepted.
- n=0 gives the empty string ε, so ε is also called accepted.

Let $M = \{Q, \Sigma, T, \delta, q_0, Z_0, F\}$

where
- $Q = \{q_0, q_1, q_f\}$
- $\Sigma = \{a, b\}$
- $T = A, Z_0$
- Initial state : q_0
- Initial stack symbol : Z_0
- $F = q_f$

Transitions

For reading a and pushing:

$\delta(q_0, a, Z_0) = \{(q_0, AZ_0)\}$

$\delta(q_0, a, A\} = \{(q_0, AA)\}$

When b starts, POP one A:

$\delta(q_0, b, A\} = \{(q_1, \varepsilon)\}$

Continue popping:-

$\delta(q_0, b, A\} = \{(q_1, \varepsilon)\}$

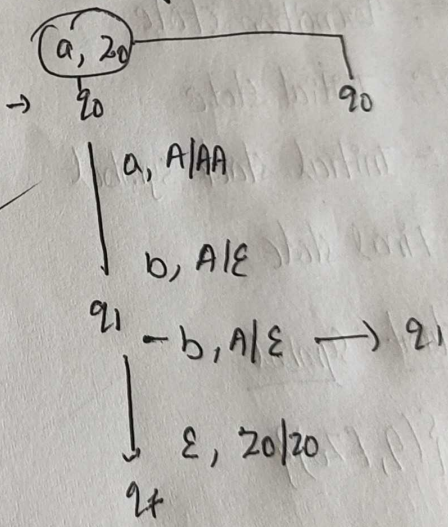
When All a's are removed:

$\delta(q_1, \varepsilon, Z_0) = \{q_f, Z_0)\}$

For n=0:

$\delta(q_0, \varepsilon, Z_0) = \{q_f, Z_0\}$

State diagram:



2. Convert CFG to PDA

Given:

$E \to E+E \mid E*E \mid (E) \mid I$

$I \to Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

we construct a PDA using the standard CFG-to-PDA conversion

Logic:-

1. Starts with the start variable E on the stack.
2. If the top of stack is a variable, replace it using a grammar production.
3. If the top is terminal, match it with the input symbol and pop it.
4. Accept when the entire input and stack are consumed.

PDA definition

$M = \{Q, \Sigma, \Gamma, \delta, q_0, Z_0, F\}$

where $Q = \{q_0, q, q_f\}$

$\Sigma = \{a, b, 0, 1, +, *, (,)\}$

Γ = Transition state

q_0 :- Initial state

Z_0 :- Initial stack symbol

F = Final state

Step 1:- Push start symbol

$\delta(q_0, \varepsilon, Z_0) = \{(q, EZ_0)\}$

Step 2: Replace variables according to productions

For E: $(q, E+E)$

$\delta(q, \varepsilon, E) = \{(q, (E)), (q, I)\}$

Step 3: Match terminals

For every terminal x:

$\delta(q, x, x) = \{(q, \varepsilon)\}$

where $x \in \{a, b, 0, 1, +, *, (,)\}$

Step 4: Accept only Z_0 remains

$\delta(q, \varepsilon, Z_0) = \{(q_f, Z_0)\}$

Thus, the PDA recognizes exactly the language generated by

Given :- $L = \{0^n 1^n / n \geq 1\}$

Logic : The TM repeatedly matches one 0 with one 1.

Use markers : X = processed 0
Y = processed 1

Definition :

$M = \{Q, \Sigma, T, \delta, q_0, B, F\}$

where $Q = \{q_0, q_1, q_2, q_3, q_4, q_5\}$
$\Sigma = \{0, 1\}$
$T = \{0, 1, X, Y, B\}$
q_0 = starting state
B = Blank
F = Final state

Transitions :

$\delta(q_0, 0) = (q_1, X, R)$

Skip processed zeros :

$\delta(q_0, X) = (q_0, X, R)$

Find matching 1 :

$\delta(q_0, 0) = (q_1, 0, R)$
$\delta(q_1, Y) = (q_1, Y, R)$
$\delta(q_1, 1) = (q_2, Y, L)$

Return left :

$\delta(q_2, 0) = (q_2, 0, L)$

$\delta(q_2, X) = (q_1, X, L)$
$\delta(q_2, Y) = (q_2, Y, L)$
$\delta(q_2, B) = (q_0, B, R)$

Accept when everything is matched

$\delta(q_0, Y) = (q_0, Y, R)$
$\delta(q_0, B) = (q_f, B, R)$

∴ The Accepted strings

01
0011
000111
00001111
..

4. Given:- $L = \{a^n b^{2n} \mid n \geq 1\}$

Here the no of b's must be twice the number of a's

Logic:-

For every a, Push two stack symbols.

Then, for every b, POP one stack symbol.

Ex: aaabbbbbb

There are 3 a's and 6 b's

Stack after reading aaa:

AAAAAA

Each b removes one A.

PDA definition:

$M = \{Q, \Sigma, T, \delta, q_0, Z_0, F\}$

where $Q = \{q_0, q_1, q_f\}$

$\Sigma = \{a, b\}$

T = Transition state $\{A, Z_0\}$

F = Final state $\{q_f\}$

Transitions:-

For each a, Push two As:

$\delta(q_0, a, Z_0) = \{(q_0, AAZ_0)\}$

$\delta(q_0, a, A) = \{(q_0, AAA)\}$

Next transition adds two more A's because one existing A is replaced by three A's

$\delta(q_1, b, A) = \{($

when stack reaches bottom:

$\delta(q_1, b, A) = \{q_2$

Ex:

aabbbbb

After aa:

AAAA

A ∴ AAAAbb

Then

$b \to AAA$

$b \to AA$

$b \to A$

$b \to Z_0$

∴ It is accepted

This

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Given: $L = \{a^n b^n c^n \mid n \geq 1\}$

This language requires equal numbers of a, b, and c

A PDA cannot generally recognize the language, but a Turing Machine can.

Logic:

- X for processed a
- Y for processed b
- Z for processed c

Definition:

$M = \{Q, \Sigma, T, \delta, q_0, B, F\}$

$Q = \{q_0, q_1, q_2, q_3, q_4, q_f, q_r\}$

$\Sigma = \{a, b, c\}$

T = Transition state

$\{a, b, c, X, Y, Z, B\}$

q_0 = Initial state

F = Final state

Transition

$\delta(q_0, X) = (q_0, X, R)$

$\delta(q_0, a) = (q_1, X, R)$

Find corresponding b.

$\delta(q_1, a) = (q, a, R)$

$\delta(q_1, Y) = (q, Y, R)$

$\delta(q_1, b) = (q_2, Y, R)$

Find corresponding c:

$\delta(q_2, b) = (q_2, b, R)$

$\delta(q_2, Y) = (q_2, Y, R)$

when c is found.

$\delta(q_2, c) = (q_3, Z, L)$

Return to beginning

In q_3, move left

when blank is reached

$\delta(q_3, B) = \{q_0, B, R\}$

Final checking

when no unmarked a remains,

the tape.

If only X, Y, Z

remain, accept.

$\delta(q_0, B) = \{q_f, B, R\}$

If an unmatched b

reject.

Ex:

aaabbbccc

TM:

aaabbbccc
Xaa Ybbzcc
XXa YYb zzc
XXX YYY zzz

Since each a, b and c has been matched exactly once.

∴ Accepted.

SIMATS ENGINEERING
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...S FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

92/100